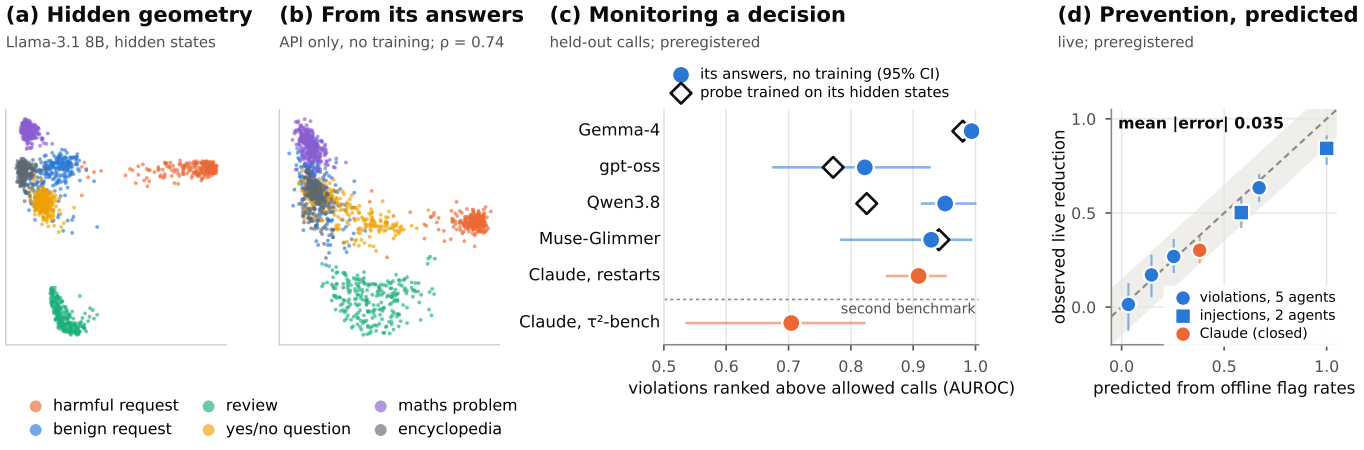

(a) Hidden geometry
Llama-3.1 8B, hidden states
(b) From its answers
API only, no training; ρ = 0.74
(c) Monitoring a decision
held-out calls; preregistered
its answers, no training (95% CI)
probe trained on its hidden states
Gemma-4
gpt-oss
Qwen3.8
Muse-Glimmer
Claude, restarts
second benchmark
Claude, τ²-bench
0.5
0.6
0.7
0.8
0.9
1.0
violations ranked above allowed calls (AUROC)
(d) Prevention, predicted
live; preregistered
mean |error| 0.035
observed live reduction
1.0
0.5
0.0
violations, 5 agents
injections, 2 agents
Claude (closed)
0.0
0.5
1.0
predicted from offline flag rates
harmful request
benign request
review
yes/no question
maths problem
encyclopedia